ECE153A Midterm
Autumn Quarter
November 6, 2001
UC Santa Barbara
Total 70 pts

I. (25 pts) For each question, choose the letter (a), (b), (c), (d), or (e) as your answer. Score = (5× number right) - number wrong, so random guessing nets you nothing on the average. If you write down the steps that arrive at your solution to a problem, you may receive partial credit.

1: You are given the following three periodic tasks. Identify a feasible real-time schedule that maximizes the total importance factor.

Task i	E_i (ms)	T_i (ms)	*Importance Factor*
1	30	60	20
2	100	200	40
3	200	400	30

(a) Task 2 (b) Tasks 1 and 2 (c) Tasks 2 and 3 (d) Tasks 1 and 3 (e) Tasks 1, 2 and 3.

2: Continue with the previous question. Suppose context switch overhead is zero and all tasks do not perform IO. What is the minimum required CPU speedup so that all three tasks are schedulable. (a) 20% (b) 25% (c) 30% (d) 40% (e) 50%.

3: Which of the following statement(s) is (are) false?

1) Interrupt driven IO is less efficient than pooled IO because the CPU is constantly interrupted by IO devices.

2) Shortest Seek Time First (SSTF) is a disk scheduling policy that can achieve the theoretical minimum average seek distance for the disk arm.

3) The sole objective of a real-time scheduler is to minimize response time.

(a) 1 and 3 (b) 2 (c) 3 (d) 1, 2, and 3 (e) 2 and 3.

Suppose a disk drive has a uniform recording density on all tracks and it has the following performance parameters. Answer the following two questions. (Hint: Because of the uniform recording density, a longer track has a higher recording capacity.)

Parameter Name	*Value*
Disk Capacity	*20 GBytes*
Number of cylinders, CYL	*10,000*
Min. Transfer Rate TR	*100 Mbps*
Max. Transfer Rate TR	*200 Mbps*
RPM	*10,000*
Min. Seek Time	*3 milliseconds*
Max. Seek Time	*15 milliseconds*

4: What is a rotational delay? (Hint: A rotational delay is the time it takes for the disk makes one full rotation.)

(a) 8.33 ms (b) 6.67 ms (c) 6 ms (d) 3 ms (e) 3.33 ms.

5: What is the capacity of the disk's outer-most cylinder? (a) 2M Bytes (b) 1.33M Bytes (c) 1M Bytes (d) 2.33M Bytes (e) 2.67M Bytes.

II. (20 pts) **True or False questions**. Indicate whether each of the following statement is *TRUE* or *FALSE*. If you choose *FALSE*, you must explain the reason(s) or no credit is given.

1: It is a good hardward/software design practice to always separate policies from mechanisms. This way, a system is much easier to adapt to a new set of performance requirements by changing its policies.

2: A harware interrupt is asynchronous in nature and it may not relate to the context of the current running process.

3 A process context switch happens when a process is moved from the *Blocked* state to the *Running* state.

4: Suppose context switch is NOT free. The average turnaround time of a non-preemptive CPU scheduling policy tends to be longer than that of a preemptive CPU scheduling policy.

5: Many (user thread) to one (kernel thread) mapping is efficient because when one user thread performs a synchronous IO, the kernel thread is not blocked and hence one of the other user threads can continue to run on the kernel thread.

III. (25 pts) Short Answers.

1 (System Calls (5 pts):
Why does a system call (such as read/write calls) need to copy arugments from the user address space to the kernel address space?

2 (Real-time Systems (10 pts): You are given the following four periodic tasks. Identify a real-time schedule that maximizes the CPU utilization.

Task i	E_i (ms)	T_i (ms)
1	40	200
2	20	100
3	20	50
4	45	300

3 (Multi-level Feedback Scheduling) (10 pts):
Suppose a system employs a Multi-Level Feedback policy with the following three scheduling rules:

1. There are three service queues in the system, each uses a round-robin scheduling policy. The first queue (the highest priority queue) has a quantum of 30 ms. The second queue has a quantum of 60 ms. The third queue (the lowest priority queue) has a quantum of 100 ms. A higher priority process always preempts a lower priority process. A preempted process is always given its full quantum back. A context switch takes 10 ms.

2. A newly arrived process is placed in the second queue.

3. A process' priority is demoted after a quantum-end and is promoted after an IO completion.

Suppose two processes, P_1 and P_2, arrive at time 0, and 10 ms, respectively, and they have the following CPU and IO bursts:

- P_1: CPU burst: 200 ms (only one burst).
- P_2: CPU burst: 40 ms; IO burst 30 ms; CPU burst: 40 ms (total three bursts).

Answer the following two questions:
(a) What is the turnaround time of each process?

(b) Propose a change to the scheduling policy so that process P_1 can complete ahead of P_2?